

Bitcoin jump clustering and the variance risk premium

Does a running jump cluster tell you the premium is about to compress?

1,838,880 one-minute bars, Jan 2023 to Jun 2026 · Deribit DVOL · code and derivations in the repository

Bitcoin's price jumps arrive in bunches rather than evenly, which is easy enough to see by eye. What I wanted to know is whether that buys you anything: if a bunch is happening right now, does it say the options market is about to get cheaper? The trouble is that “how agitated the jump process currently is” is not something you observe. So most of the work here is building a model that can estimate it, checking that the model actually works, and only then asking the question I started with.

1. What I found

Jumps cluster, and the standard model has no room for it

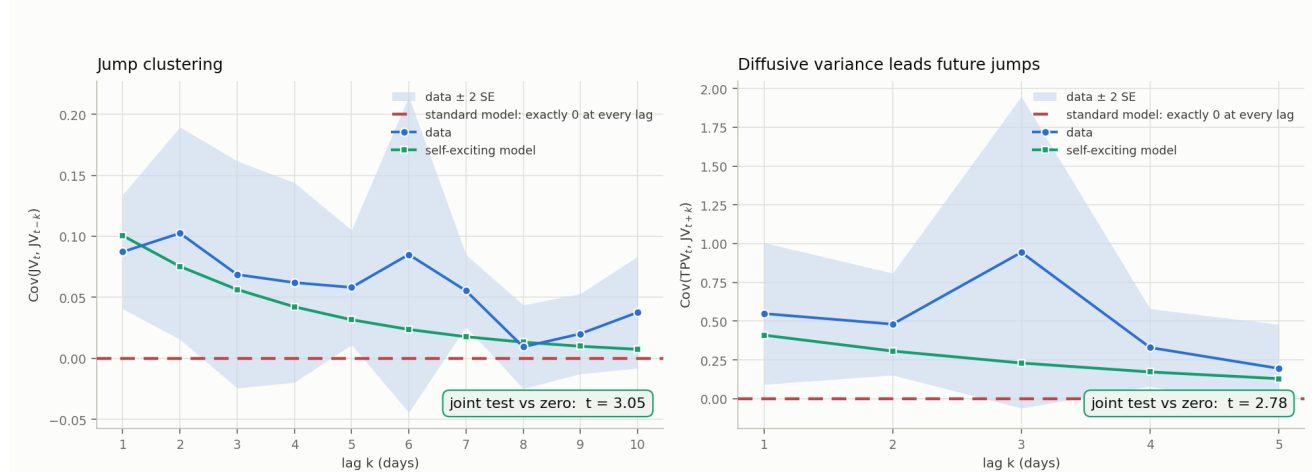
The usual jump-diffusion setup has jumps arriving at a constant rate. That carries a consequence you can check directly: jump energy on separate days is independent, so the autocovariance of daily jump variance is zero at every lag, for any parameter values whatsoever. There is nothing to tune.

It is not zero.

test	constant-intensity model says	data
Jump clustering, $\text{Cov}(JV_t, JV_{t-k})$, joint over 10 lags	0	$t = 3.05$
Diffusive variance leading future jumps, joint over 5 lags	0	$t = 2.78$
Lag-1 jump clustering on its own	0	$t = 3.74$

Bitcoin jumps cluster — the standard model says they cannot

Constant-intensity jump-diffusion predicts both curves are identically zero for every parameter value. Both are significantly positive.



Letting jumps excite themselves fixes it

The repair is to make the arrival rate depend on the jump-variance state, $\lambda(t) = \lambda_0 + \lambda_1 V^j(t^-)$, so a jump raises the odds of the next one. Both curves above come back. Fitting by GMM on 40 moments, 2 of the 40 miss by more than two standard errors, which is roughly what noise alone gives you, and the misses are scattered instead of piling up in one block.

The estimates put about 24% of jumps down as aftershocks of earlier jumps. A single shock fades with a 1.8 day half-life, but a whole cluster takes 2.4 days, which is the point of the extension: the cascade outlives the shock that started it.

The premium is large

Implied variance sits around 20% above what the model expects physically, on 65% of days, and about 25% above what actually gets realised. It mean-reverts with a half-life of roughly two weeks.

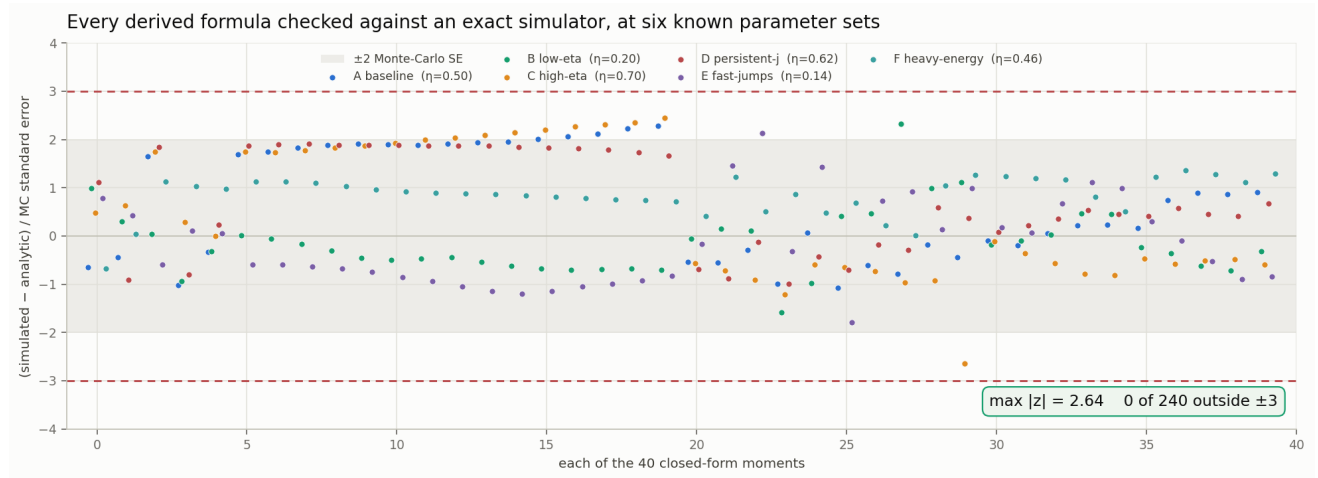
The model still decays too fast

On held-out crash events, variance stays elevated about 9.5 days. My model says 2.4. A separate over-identification check disagrees with the model in the same direction (by a factor of 6.3), so I think something slow is genuinely missing, most likely a third variance factor. The self-exciting term moves things the right way but not nearly far enough.

2. Checking the machinery before I used it

This is where most of the time went. The 40 moments are formulas I derived by hand, and if one of them is wrong then everything downstream is quietly wrong with it, in a way that no amount of staring at the output would reveal.

So before running anything on real data I wrote an exact simulator: Ogata thinning for the self-exciting jumps, the noncentral χ^2 transition for the CIR factor, and analytic integrals within each day, so there is no discretisation error anywhere to hide behind. Then I generated 50,000 days from each of six known parameter sets and checked every formula against what the simulation actually produced.



Everything landed inside Monte Carlo error, with a maximum $|z|$ of 2.64 and none of the 240 outside ± 3 . GMM recovered the true parameters, and the bootstrap intervals covered at 0.86 to 0.98 against a nominal 0.95.

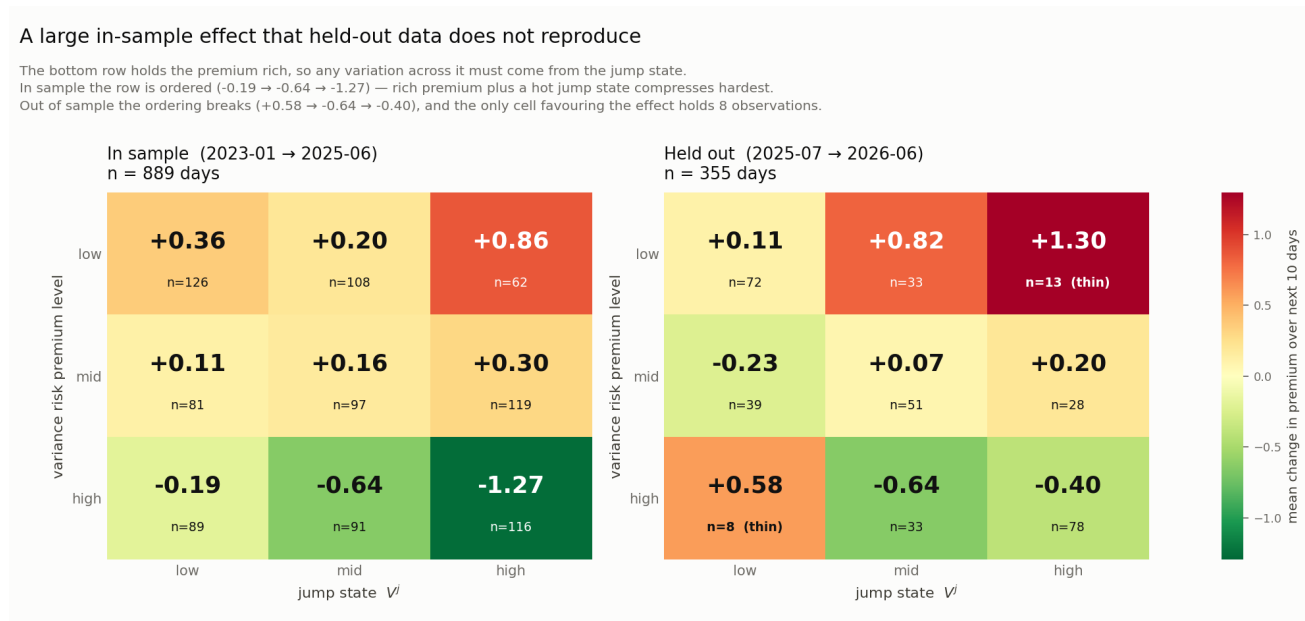
Doing this caught four real bugs:

- My tripower variation used exponent $4/3$, which estimates quarticity, when variance needs $2/3$. The “jump variance” series was effectively total variance. I found it by cross-checking against MedRV, a second jump-robust estimator, which now agrees at $\rho = 0.996$.

- The block bootstrap was resampling the raw series and recomputing lag products on the result, which splices unrelated days together at every block join. Coverage for one parameter came out at 0.00. Resampling the moment contributions instead put it back to 0.93.
- The GMM weighting matrix was shrinking toward a scaled identity, which ignores that moment variances here span five orders of magnitude, so the jump-clustering moments got almost no weight. This had inverted the persistence estimate, 3.43 against the correct 0.29, and I only noticed from the fit residuals.
- One of my two over-identification tests turned out to be an algebraic tautology. It holds for any parameter values at all, so it tested nothing, and I had already written it up as a pass. The giveaway was that it came out exactly zero with exactly zero variance in every bootstrap draw, and real quantities wobble.

3. The question I started with, and the answer

Sorting days by how rich the premium is and how hot the jump state is gives a clean pattern in sample. Holding the premium rich, a hot jump state compresses it roughly 20 times harder over the next ten days than a quiet one does, and raises the odds of a large compression by about 2.1 times.



It does not hold up out of sample. On the held-out year the ordering breaks, and the one cell that would support the effect has 8 observations in it. With something like 35 independent ten-day windows the holdout cannot really settle the question either way, but it certainly did not confirm anything.

Three other results looked good and did not survive being tested properly:

- The premium appeared to predict variance-selling returns at $t = 5.7$. But the signal and the payoff both contain the implied-variance leg, correlated at 0.97, and controlling for it leaves $t = 0.99$.
- The continuous-variance state appeared to predict premium changes at $t = 2.5$. But the model's forecast is constructed out of that state, so the relationship is mechanical, $t = 8.5$ through that channel alone. Against pure market data it is $t = 0.10$.
- The structural forecast does not beat a three-term HAR regression, R^2 of 0.117 against 0.113.

I would rather write all of this down than quietly keep whichever version looked best.

4. How it works

Total variance comes from two latent factors: a persistent Brownian-driven CIR component, and a jump-driven component whose arrival intensity is affine in its own level. Affinity is what matters, because it keeps all 40 unconditional moments in closed form. Those are the means, variances, autocovariances and two-sided cross-covariances of daily tripower variation and jump variance, and I match them to sample moments by GMM using an analytic 40×9 Jacobian, regularised weighting, and a moving-block bootstrap for intervals.

Parameters come from 2023-01 to 2025-06 only. Everything from 2025-07 onward, which includes the October 2025 crash, is held out and never touched during estimation.

5. What I would do next

The persistence gap, 9.5 days observed against 2.4 predicted, and the factor-of-6.3 over-identification gap both point at the same missing piece, so a third and slower variance factor is the obvious thing to try. The compression result needs more than one year of held-out data before it means anything either way. At this point the sample size is the constraint, not the method.

Everything reproduces with `python run_all.py`. The derivations are in `docs/affine_intensity_theory.pdf`.